

AVIRAL JAIN

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EDUCATION

Uppsala University

Master's in Data Science – Machine Learning and Statistics

Jun 2026

Uppsala, Sweden

Shri Vaishnav Vidyapeeth Vishwavidyalaya

Bachelor of Technology in Computer Science

Jun 2023

Indore, India

WORK EXPERIENCE

Summer Worker – Data Science / Machine Learning

Jun 2026 – Present

Alleima

Sandviken, Sweden

- Developed and extended computer vision and deep learning pipelines for real-world SEM and LOM production datasets, improving robustness across varying image conditions.
- Built a machine learning-based visual quality inspection system for production images, reducing manual surface inspection by 99%.
- Developed automated image analysis workflows for grain-boundary detection, phase identification and quantitative feature extraction using PyTorch, OpenCV and classical computer vision.
- Worked with production-scale datasets and transformed research prototypes into more robust, maintainable AI solutions for industrial use.

Jr. Data Scientist

Aug 2023 – Jun 2024

Moreyears

Indore, India

- Developed and deployed AI/ML applications for document processing, conversational AI and information extraction using Python, NLP, OCR and deep learning.
- Built LLM-powered applications and Retrieval-Augmented Generation (RAG) pipelines, combining language models with retrieval systems to generate context-aware responses.
- Developed production-ready AI services using Python, FastAPI and Flask, integrating machine learning models, LLM services, databases and REST APIs.
- Achieved 95%+ extraction accuracy on document processing tasks by combining **OCR, NLP and machine learning** techniques.
- Deployed AI applications on Google Cloud Platform (GCP) and worked with Docker and MLflow for deployment, experiment tracking and model lifecycle management.

PROJECTS

Master's Thesis – Quantification of Micro-structural Data using ML

Alleima

Python, PyTorch, OpenCV, Deep Learning

Jan 2026 – May 2026

- Built an automated deep learning and computer vision pipeline for grain-boundary segmentation and quantification of SEM images, achieving a Dice score of 0.80 and reducing manual quantification effort by 95%+.
- Benchmarked TransUNet, transfer learning, Random Forest, DBSCAN and morphology-based methods to evaluate accuracy and robustness across different approaches.
- Improved model robustness across variable image quality using adaptive preprocessing techniques including Otsu thresholding, Canny/Hough transforms and contrast enhancement.

AI Support in Vehicle Comfort Testing and Analysis

Scania Group

Python, GenAI, LLMs, XGBoost, SHAP, RAG

Sep 2025 – Jan 2026

- Built an XGBoost model to identify which truck components most influence driver comfort and ride quality, using SHAP to generate per-component explainability.
- Fed SHAP-derived explanations into an LLM-powered, RAG-based conversational assistant, letting engineers describe their requirements in natural language and receive AI-recommended components and add-ons best suited to their needs.
- Combined ML-based feature attribution with LLM reasoning to translate complex ride-quality data into actionable, conversational recommendations for technical users.
- Accelerated interpretation of test data across multiple truck configurations by more than 95%.

GenAI Interview & Resume Optimization Assistant

Python, LLMs, RAG, LangChain, FastAPI, GCP

- Built an LLM-powered assistant that analyzes a candidate's resume and job description to generate personalized interview questions and interactive interview practice.
- Developed an automated resume optimization workflow that identifies skill and experience gaps against target job descriptions and generates ATS-oriented improvements.
- Designed the application as an end-to-end AI workflow combining prompt engineering, document processing, LLM inference and structured output generation.
- Deployed the application end-to-end on Google Cloud Platform (GCP) using FastAPI for scalable serving of the resume analysis and interview simulation workflows.

SKILLS

- **Generative AI:** LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Conversational AI, AI Agents
- **LLM Frameworks & Tools:** LangChain, Vector Search, FAISS, ChromaDB
- **Machine Learning:** Machine Learning, Deep Learning, NLP, Computer Vision, CNNs, Transformers, Transfer Learning, SHAP, XGBoost
- **Frameworks:** PyTorch, Scikit-learn, OpenCV, NumPy, Pandas
- **Programming:** Python, SQL, R
- **AI Engineering:** FastAPI, Flask, REST APIs, Docker, Git, Linux
- **MLOps & Cloud:** MLflow, GCP, Azure, AWS SageMaker, Model Deployment
- **Databases & Data:** MySQL, MsSQL, Vector Databases, Data Processing, Feature Engineering